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(54) **CONNECTOR**

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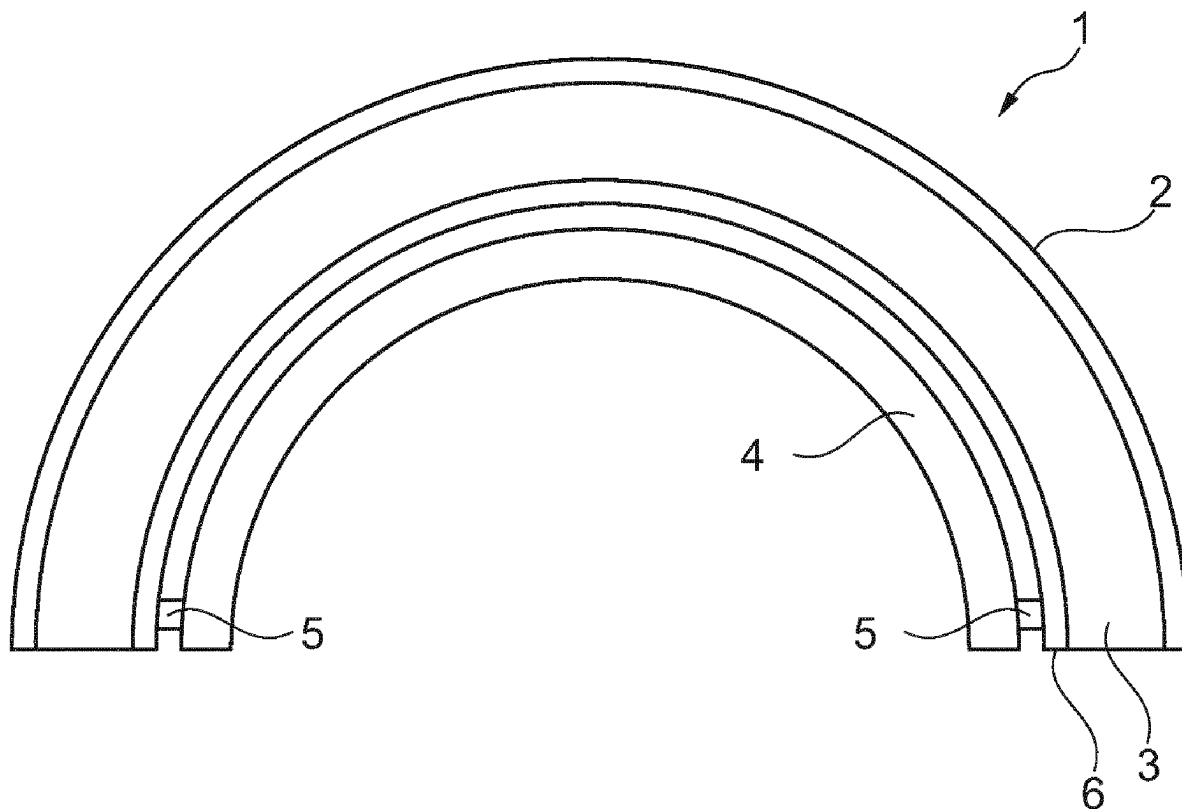
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(57)

ABSTRACT

According to a first aspect of the disclosure there is provided a connector (20) for connecting first and second parts of an apparatus, the connector comprising: a first attachment part (21) arranged on a first side of the connector, configured to be attached to the first part of the apparatus such that the first side of the connector faces in a first direction towards the first part of the apparatus: a first low friction layer (22) arranged on the first side of the connector and configured to reduce friction between the connector and the first part of the apparatus: a deformable layer (23) formed from a deformable material, provided at least between the first attachment part and the first low friction layer extending in a direction substantially perpendicularly to the first direction and configured to allow the first attachment part and first low friction layer to move relative to each other in a direction substantially perpendicularly to the first direction.



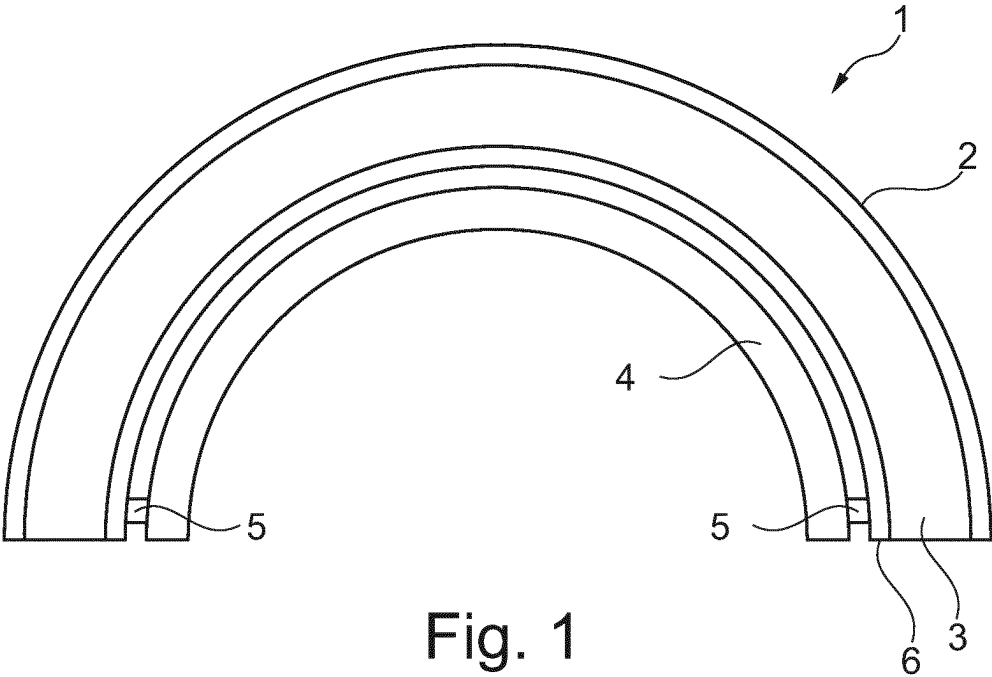


Fig. 1

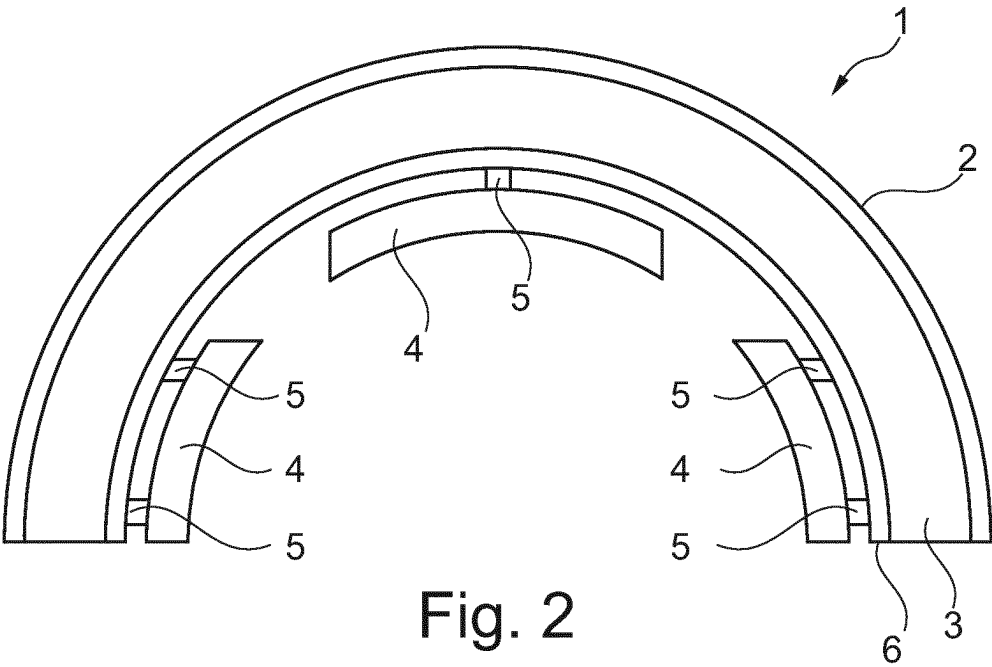


Fig. 2

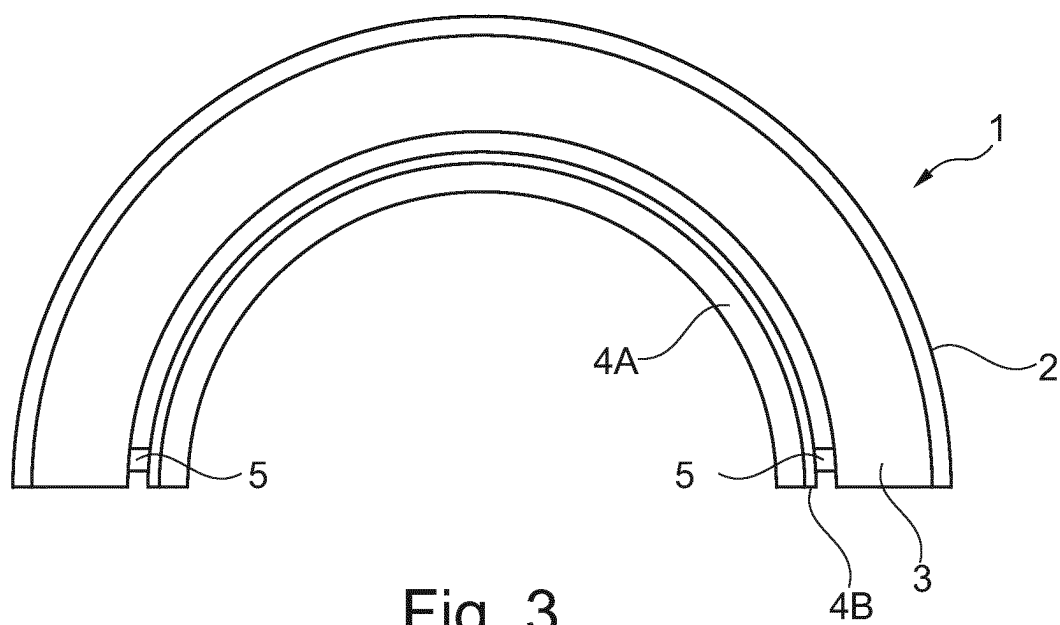


Fig. 3

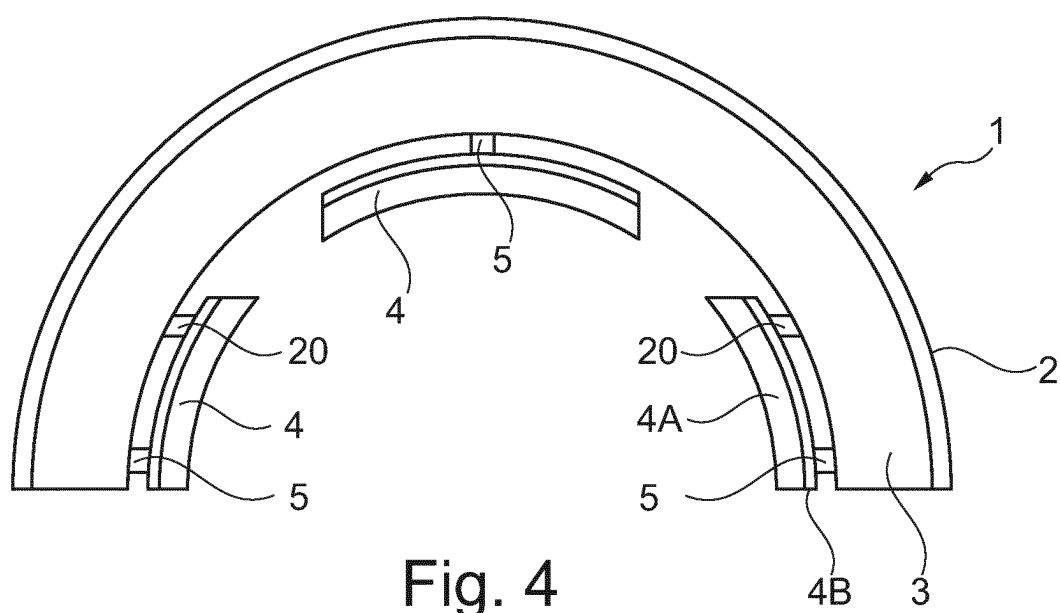


Fig. 4

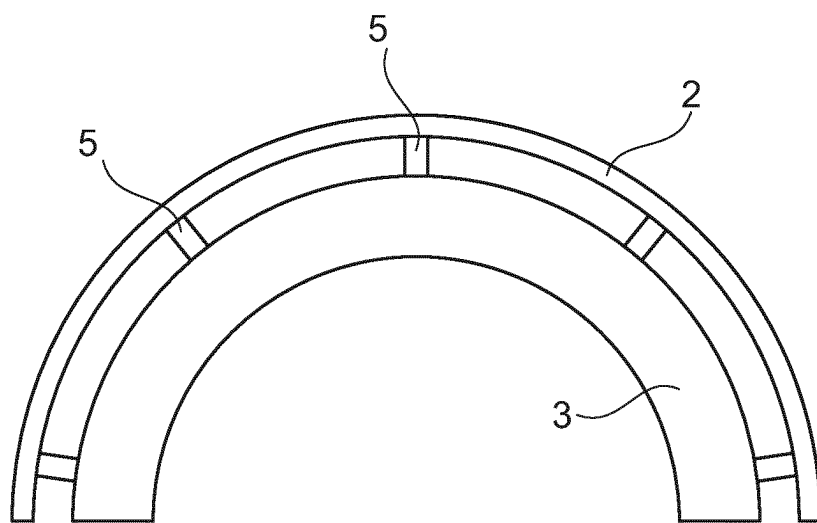


Fig. 5

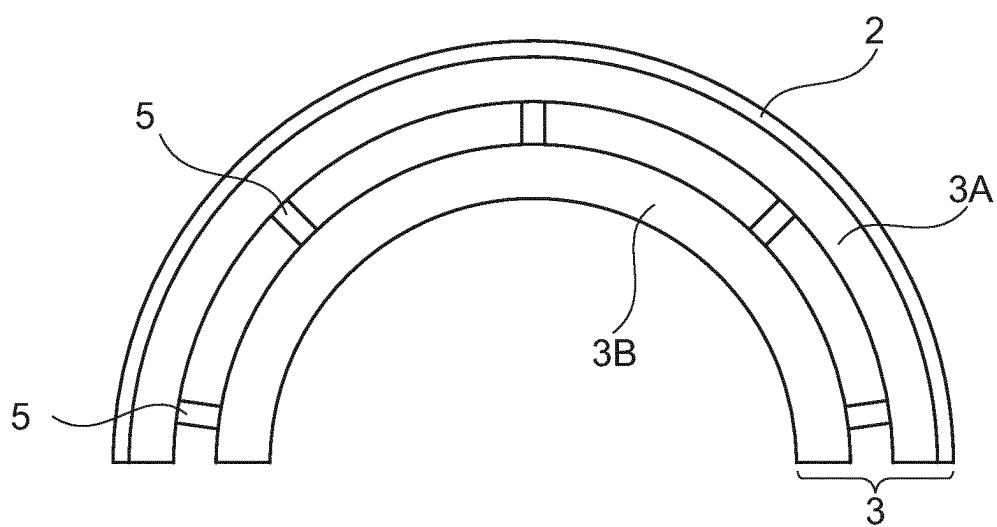


Fig. 6

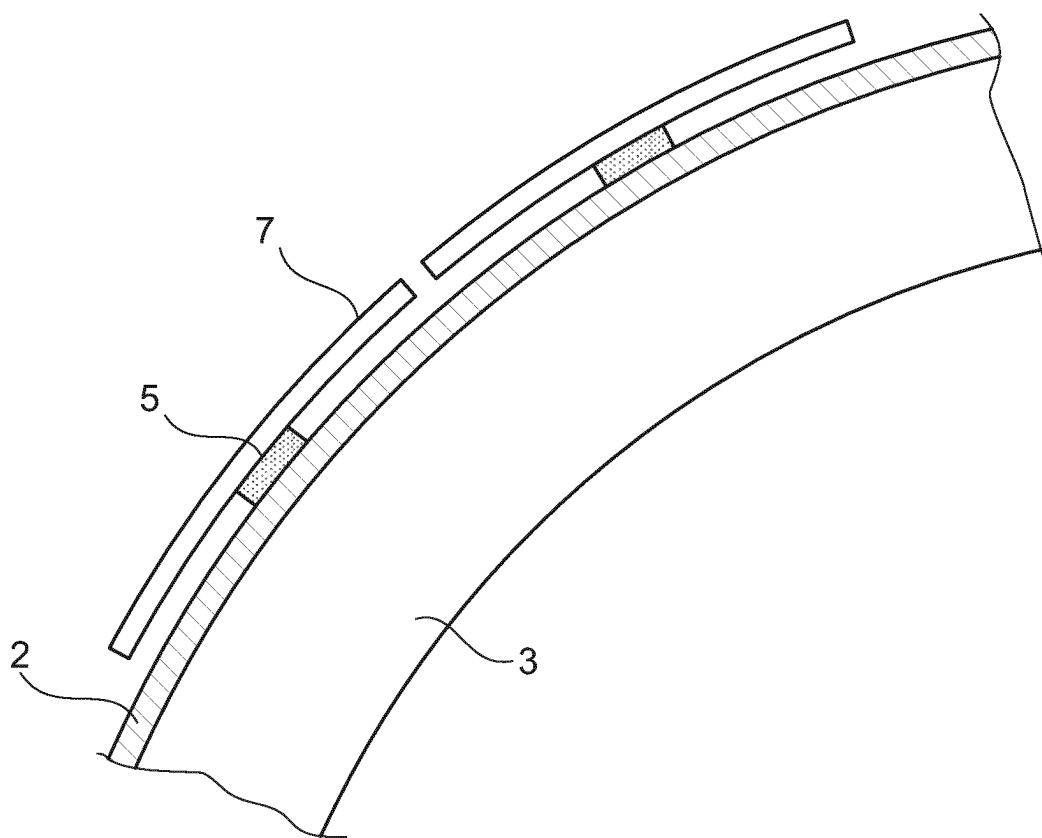


Fig. 7

Fig. 8

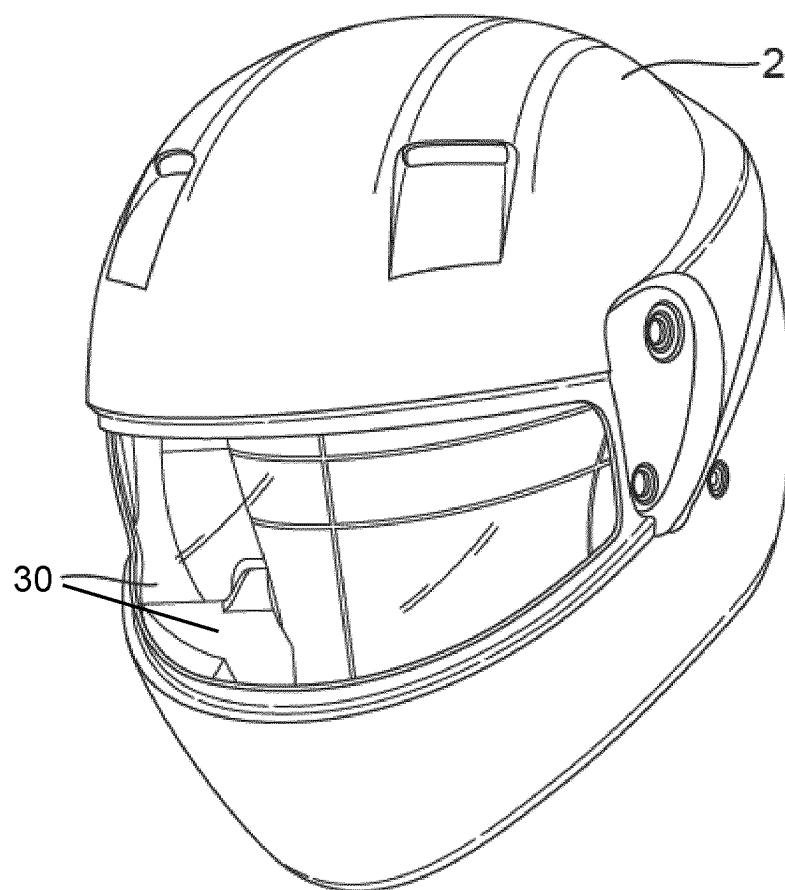


Fig. 9

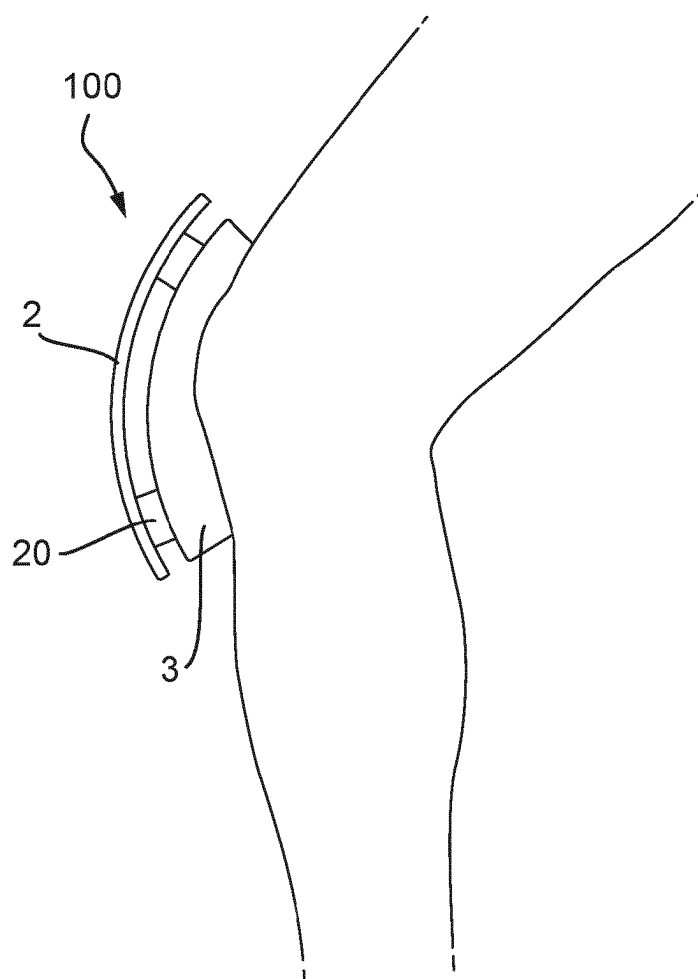


Fig. 10

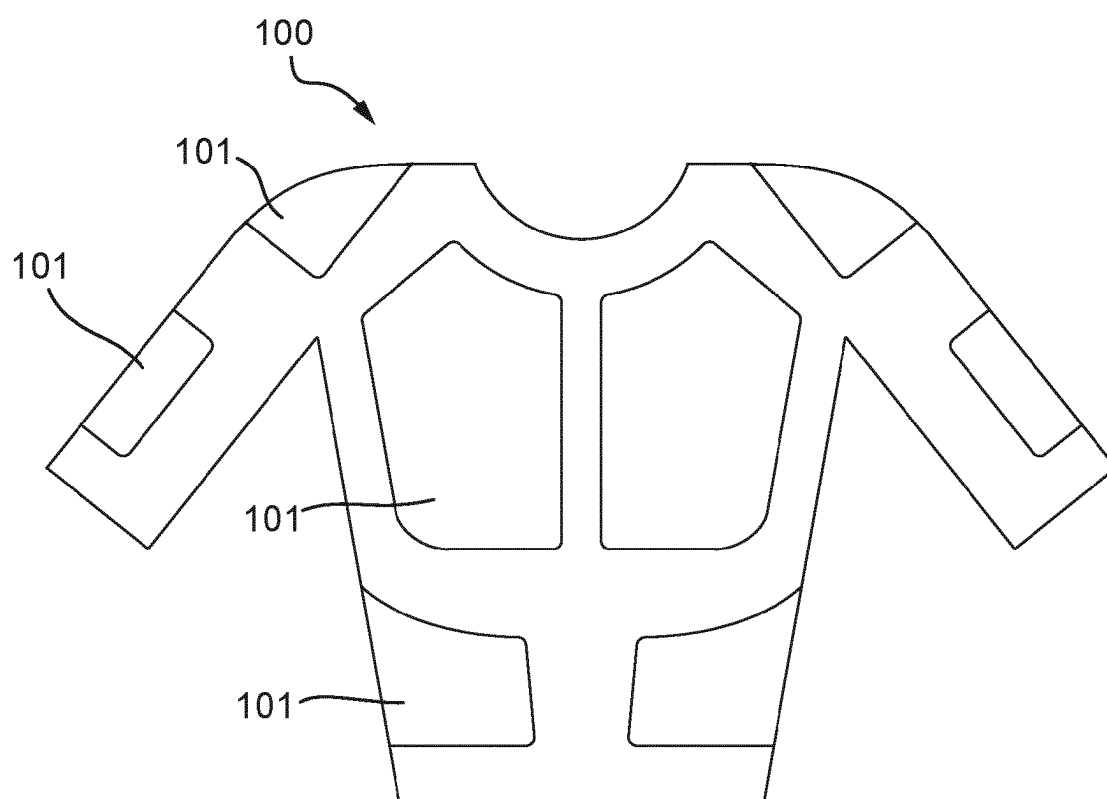


Fig. 11

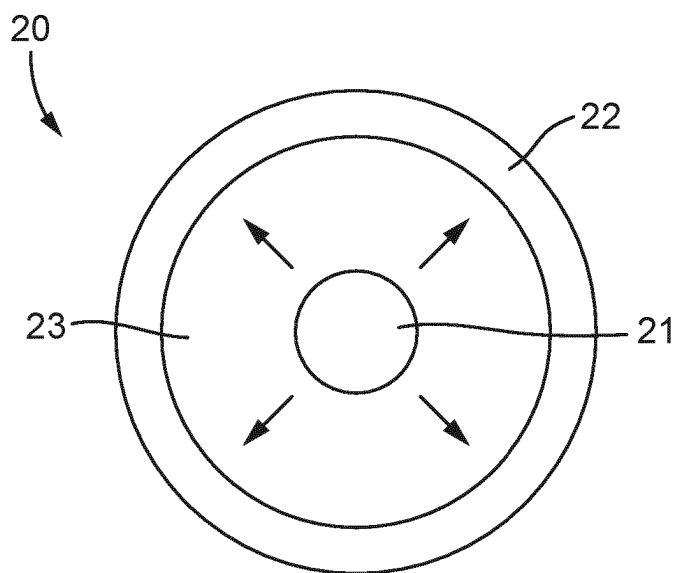


Fig. 12

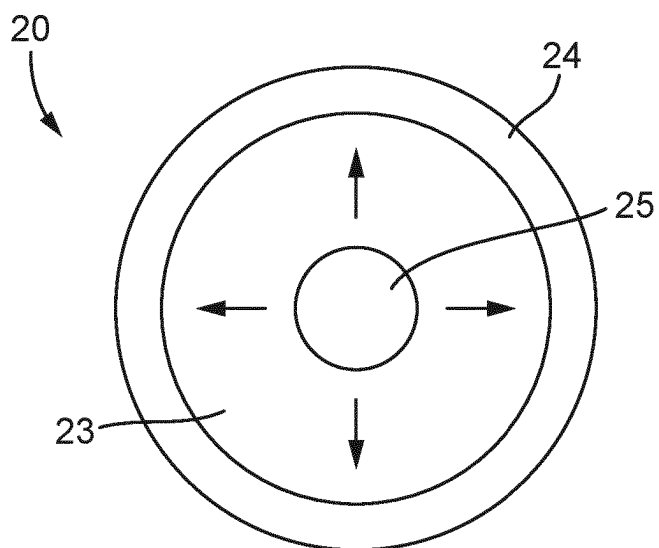


Fig. 13

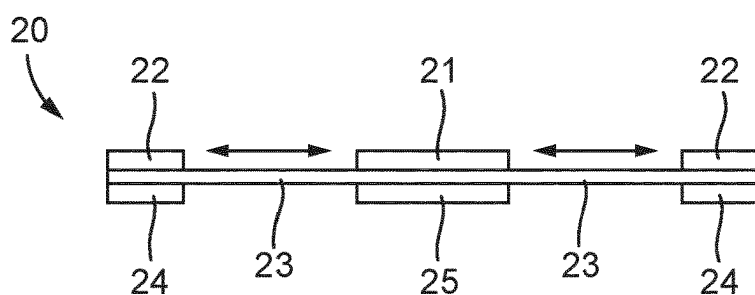


Fig. 14

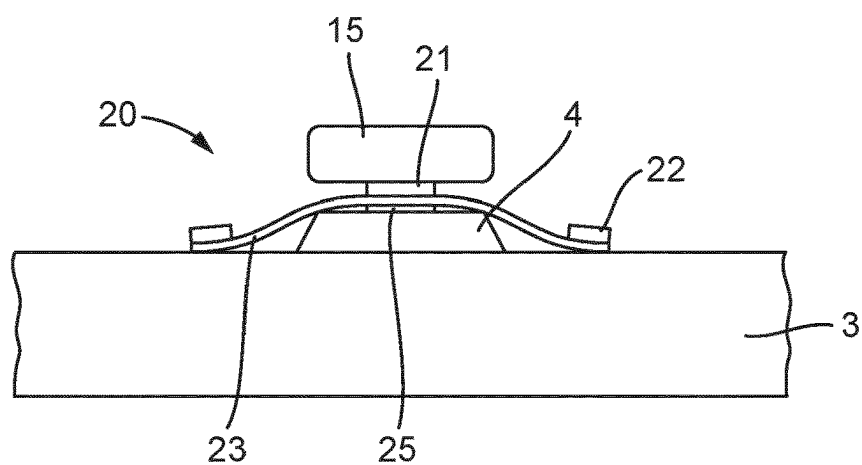
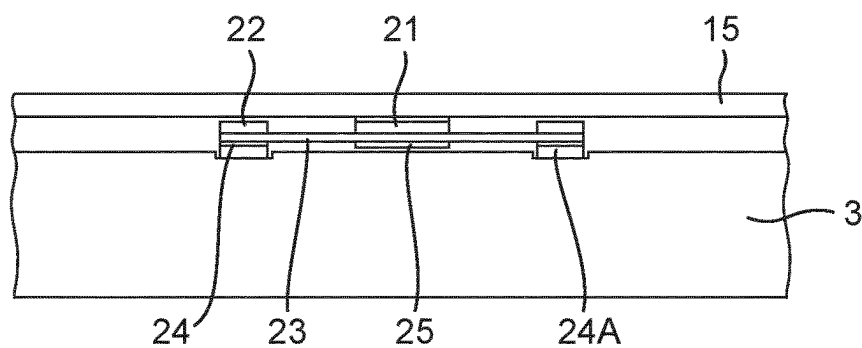


Fig. 15



CONNECTOR

TECHNICAL FIELD

[0001] The present disclosure relates to a connector for connecting first and second parts of a protective apparatus.

BACKGROUND ART

[0002] Impact protection apparatuses generally aim to reduce the energy transferred to an object, such as a person to be protected, by an impact. This may be achieved by energy absorbing means, energy redirecting means, or a combination thereof. Energy absorbing means may include energy absorbing materials, such as a foam materials, or structures configured to deform elastically and/or plastically in response to an impact. Energy redirecting means may include structures configured to slide, shear or otherwise move in response to an impact.

[0003] Impact protection apparatuses include protective apparel for protecting a wearer of the apparel. Protective apparel comprising energy absorbing means and/or energy redirecting means is known. For example, such means are implemented extensively in protective headgear, such as helmets.

[0004] Examples of helmets comprising energy absorbing means and energy redirecting means include WO 2001/045526 and WO 2011/139224 (the entirety of which are herein incorporated by reference). Specifically, these helmets include at least one layer formed from an energy absorbing material and at least one layer that can move relative to the head of the wearer of the helmet under an impact.

[0005] Implementing moving parts in a protective apparatus has challenges. For example, connecting two layers of an apparatus in such a way that permits enough relative movement between parts of the apparatus under an impact but maintains the structural integrity of the apparatus can be challenging. Ensuring that the connector can be manufactured and assembled relatively easily can be challenging.

[0006] Further, ensuring desired relative movement between moving parts under an impact can be challenging. Ensuring that the apparatus can be manufactured and assembled relatively easily can be challenging.

[0007] It is the aim of the present invention to provide a connector and an apparatus comprising the connector that at least partially address some of the problems discussed above.

SUMMARY OF THE INVENTION

[0008] According to a first aspect of the disclosure there is provided a connector for connecting first and second parts of an apparatus, the connector comprising: a first attachment part arranged on a first side of the connector, configured to be attached to the first part of the apparatus such that the first side of the connector faces in a first direction towards the first part of the apparatus; a first low friction layer arranged on the first side of the connector and configured to reduce friction between the connector and the first part of the apparatus; a deformable layer formed from a deformable material, provided at least between the first attachment part and the first low friction layer extending in a direction substantially perpendicularly to the first direction and configured to allow the first attachment part and first low friction

layer to move relative to each other in a direction substantially perpendicularly to the first direction.

[0009] Optionally, the deformable layer is formed from a sheet of the deformable material. Optionally, the deformable material is a sheet of stretchable fabric. Optionally, the first attachment part is arranged on a first side of the sheet of deformable material. Optionally, the first low friction layer is arranged on the first side of the sheet of deformable material.

[0010] Optionally, the low friction layer is formed from a sheet of low friction material.

[0011] Optionally, the first low friction layer is flexible.

[0012] Optionally, the first low friction layer is formed from a thin layer of plastic. Optionally, the plastic is polycarbonate.

[0013] Optionally, the first low friction layer is no more than 0.5 mm thick.

[0014] Optionally, the low friction layer surrounds the first attachment part. Optionally, the low friction layer is annular.

[0015] Optionally, the first attachment part comprises a hook and loop material. Optionally, the first attachment part comprises a disk of hook and loop material.

[0016] Optionally, the connector further comprises a second attachment part on a second side of the connector opposite the first side, configured to be attached to the second part of the apparatus. Optionally, the first and second attachment parts are displaced from each other in a direction along the deformable material. Optionally, the deformable material is configured to stretch by at least 1 cm, so as to provide at least 1 cm relative movement between the first attachment part and the second attachment part. Optionally, the second attachment part is arranged to substantially cover a portion of the second side of the connector opposite a portion of the first side of the connector at which the first low friction layer is arranged.

[0017] Optionally, comprising a second low friction layer on a second side of the connector opposite the first side, configured to reduce friction between the connector and the second part of the apparatus. Optionally, the second low friction layer is arranged to substantially cover a portion of the second side of the connector opposite a portion of the first side of the connector at which the first attachment part is arranged.

[0018] According to a second aspect of the disclosure there is provided an apparatus comprising: a first part; a second part; and at least one connector according to any one of the preceding claims connecting the first and second parts.

[0019] Optionally, the second part of the apparatus comprises a low friction layer arranged to face the one or more connectors.

[0020] Optionally, the apparatus is protective apparel and the first part of the apparatus comprises an interface layer configured to interface with a wearer.

[0021] Optionally, the second part comprises at least one of a hard shell and an energy absorbing layer.

[0022] Optionally, the apparatus is a helmet.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The invention is described in detail below, with reference to the accompanying figures, in which:

[0024] FIG. 1 schematically shows a cross-section through a first example helmet;

[0025] FIG. 2 schematically shows a cross-section through a second example helmet;

[0026] FIG. 3 schematically shows a cross-section through a third example helmet;
 [0027] FIG. 4 schematically shows a cross-section through a fourth example helmet;
 [0028] FIG. 5 schematically shows a cross-section through a fifth example helmet;
 [0029] FIG. 6 schematically shows a cross-section through a sixth example helmet;
 [0030] FIG. 7 schematically shows a cross-section through a seventh example helmet;
 [0031] FIG. 8 shows an eighth example helmet;
 [0032] FIG. 9 shows a first example of body armour;
 [0033] FIG. 10 shows a second example of body armour;
 [0034] FIG. 11 shows a first view of a first example connector;
 [0035] FIG. 12 shows a second view of the first example connector;
 [0036] FIG. 13 shows a third view of the first example connector;
 [0037] FIG. 14 shows a first example helmet comprising a connector; and
 [0038] FIG. 15 shows a second example helmet comprising a connector.

DETAILED DESCRIPTION

[0039] It should be noted that the Figures are schematic, the proportions of the thicknesses of the various layers, and/or of any gaps between layers, depicted in the Figures have been exaggerated for the sake of clarity and can of course be adapted according to need and requirements.

[0040] Although the examples described below relate to helmets, it should be understood that the invention applies generally to protective apparatuses, including other types of headgear and other protective apparel.

[0041] Protective apparatuses can be understood to have parts corresponding to the parts of the helmets described below. For example, protective apparatuses may have a layered structure corresponding to the layered structure of the described helmets.

[0042] Terms that are specific to a helmet, such as “radial direction” can be understood to have equivalents in the context of other protective equipment, such as “thickness direction”. A “wearer” is to generally understood as corresponding to an object that is to be protected by the protective apparatus, and “head” as a specific part of the object, e.g. a different body part, with which the apparatus is in contact.

[0043] General features of the example helmets are described below with reference to FIGS. 1 to 7.

[0044] FIGS. 1 to 7 show example helmets 1 comprising an energy absorbing layer 3. The purpose of the energy absorbing layer 3 is to absorb and dissipate energy from an impact in order to reduce the energy transmitted to the wearer of the helmet. Within the helmet 1, the energy absorbing layer may be the primary energy absorbing element. Although other elements of the helmet 1 may absorb that energy to a more limited extent, this is not their primary purpose.

[0045] The energy absorbing layer 3 may absorb energy from a radial component of an impact more efficiently than a tangential component of an impact. The term “radial” generally refers to a direction substantially toward the centre of the wearers head, e.g. substantially perpendicular to an outer surface of the helmet 1. The term “tangential” may

refer to a direction substantially perpendicular to the radial direction, in a plane comprising the radial direction and the impact direction.

[0046] The energy absorbing layer may be formed from an energy absorbing material, such as a foam material. Preferable such materials include expanded polystyrene (EPS), expanded polypropylene (EPP), expanded polyurethane (EPU), vinyl nitrile foam; or strain rate sensitive foams such as those marketed under the brand-names Poron™ and D3O™.

[0047] Alternatively, or additionally, the energy absorbing layer may have a structure that provides energy absorbing characteristics. For example, the energy absorbing layer may comprise deformable elements, such as cells or finger-like projections, that deform upon impact to absorb and dissipate the energy of an impact.

[0048] As illustrated in FIG. 6, the energy absorbing layer 3 of the helmet 1 is divided into outer and inner parts 3A, 3B.

[0049] The energy absorbing layer is not limited to one specific arrangement or material. The energy absorbing layer 3 may be provided by multiple layers having different arrangements, i.e. formed from different materials or having different structures. The energy absorbing layer 3 may be a relatively thick layer. For example, it may be the thickest layer of the helmet 1.

[0050] FIGS. 1 to 7 show example helmets 1 comprising an outer layer 2. The purpose of the outer layer 2 may be to provide rigidity to the helmet. This may help spread the impact energy over a larger area of the helmet 1. The outer layer 2 may also provide protection against objects that might pierce the helmet 1. Accordingly, the outer shell may be a relatively strong and/or rigid layer, e.g. compared to an energy absorbing layer 3. The outer layer 2 may be a relatively thin layer, e.g. compared to an energy absorbing layer 3.

[0051] The outer layer 2 may be formed from a relatively strong and/or rigid material. Preferable such materials include a polymer material such as polycarbonate (PC), polyvinylchloride (PVC) or acrylonitrile butadiene styrene (ABS) for example. Advantageously, the polymer material may be fibre-reinforced, using materials such as glass-fibre, Aramid, Twaron, carbon-fibre and/or Kevlar.

[0052] As shown in FIG. 7, one or more outer plates 7 may be mounted to the outer layer 2 of the helmet 1. The outer plates 7 may be formed from a relatively strong and/or rigid material, for example from the same types of materials as from which the outer layer 2 may be formed. The selection of material used to form the outer plates 7 may be the same as, or different from, the material used to form the outer layer 2.

[0053] In some example helmets, the outer layer 2 and/or the energy absorbing layer 3 may be adjustable in size in order to provide a customised fit. For example the outer layer 2 may be provided in separate front and back parts. The relative position of the front and back parts may be adjusted to change the size of the outer layer 2. In order to avoid gaps in the outer layer 2, the front and back parts may overlap. The energy absorbing layer 3 may also be provided in separate front and back parts. These may be arranged such that the relative position of the front and back parts may be adjusted to change the size of the energy absorbing layer 3. In order to avoid gaps in the energy absorbing layer 3, the front and back parts may overlap.

[0054] FIGS. 1 to 4 shows example helmets 1 comprising an interface layer 4. Although not shown in FIGS. 5 to 7, these example helmets may also comprise an interface layer 4. The purpose of interface layer 4 may be to provide an interface between the helmet and the wearer. In some arrangements, this may improve the comfort of the wearer. The interface layer 4 may be provided to mount the helmet on the head of a wearer. The interface layer 4 may be provided as a single part or in multiple sections.

[0055] The interface layer 4 may be configured to at least partially conform to the head of the wearer. For example, the interface layer 4 may be elastically and/or may comprise an adjustment mechanism for adjusting the size of the interface layer 4. In an arrangement, the interface layer may engage with the top of a wearer's head. Alternatively or additionally, the interface layer 4 may comprise an adjustable band configured to encircle the wearer's head.

[0056] The interface layer 4 may comprise comfort padding 4A. Multiple sections of comfort padding 4A may be provided. The comfort padding 4A may be provided on a substrate 4B for mounting the comfort padding to the rest of the helmet 1.

[0057] The purpose of the comfort padding 4A is to improve comfort of wearing the helmet and/or to provide a better fit. The comfort padding may be formed from a relatively soft material, e.g. compared to the energy absorbing layer 3 and/or the outer layer 2. The comfort padding 4A may be formed from a foam material. However, the foam material may be of lower density and/or thinner than foam materials used for the energy absorbing layer 3. Accordingly, the comfort padding 4A will not absorb a meaningful amount of energy during an impact, i.e. for the purposes of reducing the harm to the wearer of the helmet. Comfort padding is well recognised in the art as being distinct from energy absorbing layers, even if they may be constructed from somewhat similar materials.

[0058] The interface layer 4, and/or comfort padding 4A that may be part of it, may be removable. This may enable the interface layer 4 and/or comfort padding 4A to be cleaned and/or may enable the provision of an interface layer and/or comfort padding 4A that is configured to fit a specific wearer.

[0059] Straps, e.g. chin straps, may be provided to secure the helmet 1 to the head of the wearer.

[0060] The helmets of FIGS. 1 to 4 are configured such that the interface layer 4 is able to move, for example slide, in a tangential direction relative to the energy absorbing layer 3 in response to an impact. As shown in FIGS. 1 to 4, the helmet may also comprise connectors 5 between the energy absorbing layer 3 and the interface layer 4 that allow relative movement between the energy absorbing layer 3 and the interface layer 4 while connecting the elements of the helmet together.

[0061] The helmet of FIG. 5 is configured such that the outer layer 2 is able to move, for example slide, in a tangential direction relative to the energy absorbing layer 3 in response to an impact. As shown in FIG. 5, the helmet 1 may also comprise connectors 5 between the energy absorbing layer 3 and the outer layer 2 that allow relative movement between the energy absorbing layer 3 and the outer layer 2 while connecting the elements of the helmet together.

[0062] The helmet of FIG. 6 is configured such that the outer part 3A of the energy absorbing layer 3 is able to move, for example slide, in a tangential direction relative to the

inner part 3B of the energy absorbing layer 3 in response to an impact. As shown in FIG. 6, the helmet 1 may also comprise connectors 5 between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3, that allow relative movement between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3, while connecting the elements of the helmet together.

[0063] The helmet of FIG. 7 is configured such that the outer plates 8 are able to move, for example slide, in a tangential direction relative to the outer layer 2 in response to an impact. As shown in FIG. 7, the helmet may also comprise connectors 5 between the outer plates 8 and the outer layer 2 that allow relative movement between the outer plates 7 and the outer layer 2, while connecting the elements of the helmet together.

[0064] The purpose of helmet layers that move or slide relative to each other may be to redirect energy of an impact that would otherwise be transferred to the head the wearer. This may improve the protection afforded to the wearer against a tangential component of the impact energy. A tangential component of the impact energy would normally result in rotational acceleration of the head of the wearer. It is well known that such rotation can cause brain injury. It has been shown that helmets with layers that move relative to each other can reduce the rotational acceleration of the head of the wearer. A typical reduction may be roughly 25% but reductions as high as 90% may be possible in some instances.

[0065] Preferably, relative movement between helmet layers results in a total shift amount of at least 0.5 cm between an outermost helmet layer and an inner most helmet layer, more preferably at least 1 cm, more preferably still at least 1.5 cm. Preferably the relative movement can occur in any direction, e.g. in a circumferential direction around the helmet, left to right, front to back and any direction in between.

[0066] Relative movement can be considered to occur substantially in a plane over the relevant ranges, even though movement between layers may be rotational rather than linear. Accordingly, reference may be made below to movement in a plane.

[0067] Regardless of how helmet layers are configured to move relative to each other, it is preferable that the relative movement, such as sliding, is able to occur under forces typical of an impact for which the helmet is designed (for example an impact that is expected to be survivable for the wearer). Such forces are significantly higher than forces that a helmet may be subject to during normal use. Impact forces tend to compress layers of the helmet together, increasing the reaction force between components and thus increasing frictional forces. Where helmets are configured to have layers sliding relative to each other the interface between them may need to be configured to enable sliding even under the effect of the high reaction forces experienced between them under an impact.

[0068] As shown in FIGS. 1 to 7, a sliding interface may be provided between the layers of the helmet 1 that are configured to slide relative to each other. At the sliding interface, surfaces slide against each other to enable relative sliding between the layers of the helmet 1. The sliding interface may be a low friction interface. Accordingly, friction reducing means may be provided at the sliding

interface. Example sliding interfaces are described further below, in relation to each of the example helmets **1** shown in FIGS. **1** to **7**.

[0069] The friction reducing means may be a low friction material or lubricating material. These may be provided as a continuous layer, or multiple discrete patches, or portions of material, for example. Possible low friction materials for the friction reducing means include waxy polymers such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE, Teflon™, a woven fabric such as Tamarack™, a non-woven fabric, such a felt. Such low friction materials may have a thickness of roughly 0.1-5 mm, but other thicknesses can also be used, depending on the material selected and the performance desired. Possible lubricating materials include oils, polymers, microspheres, or powders. Combinations of the above may be used.

[0070] In one example the low friction material or lubricating material may be a polysiloxane-containing material. In particular the material may comprise (i) an organic polymer, a polysiloxane and a surfactant; (ii) an organic polymer and a copolymer based on a polysiloxane and an organic polymer; or (iii) a non-elastomeric cross-linked polymer obtained or obtainable by subjecting a polysiloxane and an organic polymer to a cross-linking reaction. Preferred options for such materials are described in WO2017/148958.

[0071] In one example the low friction material or lubricating material may comprise a mixture of (i) an olefin polymer, (ii) a lubricant, and optionally one or more further agents. Preferred options for such materials are described in WO2020/115063.

[0072] In one example the low friction material or lubricating material may comprise an ultra high molecular weight (UHMW) polymer having a density of ≤ 960 kg/m³, which UHMW polymer is preferably an olefin polymer. Preferred options for such materials are described in WO2020/115063.

[0073] In one example the low friction material or lubricating material may comprise a polyketone. Preferred options for such materials are described in WO 2020260185.

[0074] In some arrangements, it may be desirable to configure the low friction interface such that the static and/or dynamic coefficient of friction between materials forming sliding surfaces at the sliding interface is between 0.001 and 0.3 and/or below 0.15. The coefficient of friction can be tested by standard means, such as standard test method ASTM D1894.

[0075] The friction reducing means may be provided on or be an integral part of one or both of the layers of the helmet **1** that are configured to slide relative to each other. In some examples, helmet layers may have a dual function, including functioning as a friction reducing means. Alternatively, or additionally, the friction reducing means may be a separate from the layers of the helmet **1** that are configured to slide relative to each other, but provided between the layers.

[0076] Instead of the sliding interface, in some examples, a shearing interface may be provided between the layers of the helmet **1** that are configured to move relative to each other. At the shearing interface, a shearing layer shears to enable relative movement between the layers of the helmet **1**. The shearing layer may comprise a gel or liquid, which may be retained within a flexible envelope. Alternatively, the shearing layer may comprise two opposing layers connected by deformable elements that deform to enable shearing between the two opposing layers.

[0077] A single shearing layer may be provided that substantially fills the volume between two layers of a helmet. Alternatively, one or more shearing layers may be provided that fill only a portion of the volume between two layers of a helmet, e.g. leaving substantial space around the shearing layers. The space may comprise a sliding interface, as described above. As such, helmets may have a combination of shearing and sliding interfaces. Such shearing layers may act as connectors **5**, which are described further below.

[0078] FIGS. **1** to **7** schematically show connectors **5**. The connectors **5** are configured to connect two layers of the helmet while enabling relative movement, e.g. sliding or shearing, between the layers. Different numbers of connectors **5** may be provided than as shown in FIGS. **1** to **7**. The connectors **5** may be located at different positions than as shown in FIGS. **1** to **7**, for example at a peripheral edge of the helmet **1** instead of a central portion.

[0079] Typically, a connector **5** comprises first and second attachment parts respectively configured to attach to first and second parts of the helmet and a deformable part between the first and second attachment parts that enables the first and second attachment parts to move relative to each other to enable movement between the first and second parts of the helmet of the helmet. Connectors **5** may absorb some impact energy by deforming.

[0080] The specific arrangements of each of the example helmets shown in FIGS. **1** to **7** are described below.

[0081] FIG. **1** shows a helmet comprising an outer layer **2**, an energy absorbing layer **3** and an interface layer **4**. The interface layer **4** is provided as a single layer and comprises comfort padding.

[0082] The helmet of FIG. **1** is configured such that the interface layer **4** is able to slide relative to the energy absorbing layer **3** in response to an impact. A sliding interface is provided between the interface layer **4** and the energy absorbing layer **3**.

[0083] A sliding layer **7** is provided on a surface of the energy absorbing layer **3** facing the sliding interface. The sliding layer **7** may be moulded to the energy absorbing layer **3** or otherwise attached thereto. The sliding layer **7** may be formed from a relatively hard material, e.g. relative to the energy absorbing layer **3**. The sliding layer **7** is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the sliding layer **7** from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the sliding layer **7**, and/or applying a lubricant to the sliding layer **7**.

[0084] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer **3** from a low friction material, by applying a low friction coating to the energy absorbing layer **3** and/or applying a lubricant to the energy absorbing layer **3**.

[0085] The helmet **1** shown in FIG. **1** also comprises connectors **5** attached to the interface layer **4**. The connectors are also connected to the sliding layer **7** to allow relative sliding between the energy absorbing layer **3** and the interface layer **4**. Alternatively, or additionally, one or more of the connectors **5** may be connected to another part of the remainder of the helmet **1**, such as the energy absorbing

layer 3 or the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0086] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0087] FIG. 2 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a plurality of independent sections each comprising comfort padding.

[0088] The helmet of FIG. 2 is configured such that the section of the interface layer 4 are able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the sections of the interface layer 4 and the energy absorbing layer 3.

[0089] A sliding layer 7 is provided on a surface of the energy absorbing layer 3 facing the sliding interface. The sliding layer 7 may be moulded to the energy absorbing layer 3 or otherwise attached thereto. The sliding layer 7 may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The sliding layer 7 is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the sliding layer 7 from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the sliding layer 7, and/or applying a lubricant to the sliding layer 7.

[0090] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0091] The helmet 1 shown in FIG. 2 also comprises connectors 5 attached to each independent section of the interface layer 4. The connectors 5 are also attached to the sliding layer 7 to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the energy absorbing layer 3 or the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0092] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0093] FIG. 3 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a single layer and comprises comfort padding 4A attached to a substrate 4B. The substrate 4B may be bonded to the outer side of the comfort padding 4A. Such bonding could be through any means, such as by adhesive or by high frequency welding or stitching.

[0094] The helmet of FIG. 3 is configured such that the interface layer 4 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the interface layer 4 and the energy absorbing layer 3.

[0095] The substrate 4B of the interface layer 4 faces the sliding interface. The substrate 4B may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3 and/or the comfort padding 4A. The substrate 4B is configured to provide friction reducing means to reduce the

friction at the sliding interface. This may be achieved by forming the substrate 4B from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the substrate 4B, and/or applying a lubricant to the substrate 4B. In alternative example, the substrate 4B may be formed from a fabric material, optionally coated with a low friction material.

[0096] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0097] The helmet 1 shown in FIG. 3 also comprises connectors 5 attached to the interface layer 4. The connectors are also connected to the energy absorbing layer 3 to allow relative sliding between the energy absorbing layer 3 and the interface layer 4. Alternatively, or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0098] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0099] FIG. 4 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a plurality of independent sections each comprising comfort padding 4A attached to a substrate 4B. The substrate 4B may be bonded to the outer side of the comfort padding 4A. Such bonding could be through any means, such as by adhesive or by high frequency welding or stitching.

[0100] The helmet of FIG. 4 is configured such that the interface layer 4 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the interface layer 4 and the energy absorbing layer 3.

[0101] The substrate 4B of the sections of the interface layer 4 faces the sliding interface. The substrate 4B may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3 and/or the comfort padding 4A. The substrate 4B is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the substrate 4B from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the substrate 4B, and/or applying a lubricant to the substrate 4B. In alternative example, the substrate 4B may be formed from a fabric material, optionally coated with a low friction material.

[0102] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0103] The helmet 1 shown in FIG. 4 also comprises connectors 5 attached to the sections of the interface layer 4. The connectors 5 are also connected to the energy absorbing layer 3 to allow relative sliding between the energy absorb-

ing layer 3 and the interface layer 4. Alternatively, or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0104] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0105] FIG. 5 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. Although not shown, an interface layer may additionally be provided.

[0106] The helmet of FIG. 5 is configured such that the outer layer 2 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface may be provided between the outer layer 2 and the energy absorbing layer 3.

[0107] Although not shown, an additional layer may be provided on a surface of the energy absorbing layer 3 facing the sliding interface. The additional layer may be moulded to the energy absorbing layer 3 or otherwise attached thereto. The additional layer may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The additional layer may be configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the additional layer from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the additional layer and/or applying a lubricant to the additional layer.

[0108] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the outer layer 2 from a low friction material, providing an additional low friction layer on a surface of the outer layer 2 facing the sliding interface, by applying a low friction coating to the outer layer 2 and/or applying a lubricant to the outer layer 2.

[0109] The helmet 1 shown in FIG. 5 also comprises connectors 5 attached to the outer layer 2. The connectors 5 are also attached to the energy absorbing layer 3 (or additional layer) to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as an interface layer. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0110] It should be understood that such an arrangement of the outer shell 2 and the energy absorbing layer 3 may be added to any helmet described herein.

[0111] FIG. 6 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. As illustrated, the energy absorbing layer 3 of the helmet shown in FIG. 6 is divided into outer and inner parts 3A, 3B. Although not shown, an interface layer may additionally be provided.

[0112] The helmet of FIG. 6 is configured such that the outer part 3A of the energy absorbing layer 3 is able to slide relative to the inner part 3B of the energy absorbing layer 3 in response to an impact. A sliding interface may be provided between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3.

[0113] Although not shown, an additional layer may be provided on a surface of one or both of the inner and outer parts 3A, 3B of the energy absorbing layer 3 facing the sliding interface. The additional layer may be moulded to the inner or outer parts 3A, 3B of the energy absorbing layer 3 or otherwise attached thereto. The additional layer may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The additional layer may be configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the additional layer from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE.

[0114] Alternatively, or additionally, this may be achieved by applying a low friction coating to the additional layer and/or applying a lubricant to the additional layer.

[0115] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming one or both of the inner and outer parts 3A, 3B of the energy absorbing layer 3 from a low friction material, providing an additional low friction layer on a surface of the inner and outer parts 3A, 3B of the energy absorbing layer 3 facing the sliding interface, by applying a low friction coating to the inner and outer parts 3A, 3B of the energy absorbing layer 3 and/or applying a lubricant to the inner and outer parts 3A, 3B of the energy absorbing layer 3.

[0116] The helmet 1 shown in FIG. 6 also comprises connectors 5 attached to the outer layer 2.

[0117] The connectors 5 are also attached to the energy absorbing layer 3 (or additional layer) to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as an interface layer. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0118] It should be understood that such an arrangement of inner and outer parts 3A 3B of the energy absorbing layer 3 may be added to any helmet described herein.

[0119] FIG. 7 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. As shown in FIG. 7, one or more outer plates 7 are mounted to the outer layer 2 of the helmet 1. The outer plates 7 may be formed from a relatively strong and/or rigid material, for example from the same types of materials as from which the outer layer 2 may be formed. Although not shown, an interface layer may additionally be provided.

[0120] The helmet of FIG. 7 is configured such that the outer plates 8 are able to slide relative to the outer layer 2 in response to an impact. A sliding interface may be provided between the outer plates 8 and the outer layer 2.

[0121] Friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the outer layer 2 and/or the outer plates 8 from a low friction material, providing an additional low friction layer on a surface of the outer layer 2 and/or the outer plates 8 facing the sliding interface, by applying a low friction coating to the outer layer 2 and/or the outer plates 8, and/or applying a lubricant to the outer layer 2 and/or the outer plates 8.

[0122] The helmet 1 shown in FIG. 7 also comprises connectors 5 attached to the outer plates 7. The connectors 5 are also attached to the outer layer 2 to allow relative sliding between the plates 7 and the outer layer 2. Alternatively or

additionally, one or more of the connectors **5** may be connected to another part of the remainder of the helmet **1**, such as the energy absorbing layer **3**. The connectors **5** may also be connected to two or more parts of the remainder of the helmet **1**.

[0123] In such an arrangement, in the event of an impact on the helmet **1**, it can be expected that the impact would be incident on one or a limited number of the outer plates **17**. Therefore, by configuring the helmet such that the one or more outer plates **7** can move relative to the outer layer **2** and any outer plates **7** that have not been subject to an impact, the surface receiving the impact, namely one or a limited number of outer plates **7**, can move relative to the remainder of the helmet **1**. In the case of an impact, this may reduce the rotational acceleration of the head of a wearer.

[0124] It should be understood that such an arrangement of outer plates **7** may be added to any helmet described herein, namely an arrangement having a sliding interface between at least two of the layers of the helmet **1**.

[0125] Some helmets, such as those shown in FIGS. **1** to **6**, are configured to cover a top portion of the head and the above described helmet structures are appropriately located in the helmet to cover a top portion of the head. For example, a helmet may be provided to substantially cover the forehead, top of the head, back of the head and/or temples of the wearer. The helmet may substantially cover the cranium of the wearer.

[0126] Some helmets may be configured to cover other parts of the head, alternatively or additionally to a top portion. For example, helmets such as the helmet shown in FIG. **8** may cover the cheeks and/or chin of the wearer. Such helmets may be configured to substantially cover the jaw of the wearer. Helmets of the type shown in FIG. **8**, are often referred to as full-face helmets. As shown in FIG. **8**, cheek pads **30** may be provided on either side of the helmet **1** (i.e. left and right sides). The cheek pads **30** may be arranged within an outer shell **2** of the helmet **1** to protect the side of the face of the wearer from an impact.

[0127] The cheek pads **30** may have the same layered structure as the example helmets described above. For example, the cheek pads **30** may comprise one or more energy absorbing layers as described above, and/or an interface layer as described above, and/or layers that move relative to each other as described above, optionally, layers may be connected by connectors as described above. Alternatively or additionally, the cheek pads **30** themselves may be configured to move relative to the outer shell **2** and, optionally be connected to the outer shell by connectors as described above.

[0128] Although, the above examples relate to helmets, as stated above, the disclosure may also relate to alternative protective apparel, such as body armour, as shown in FIGS. **9** and **10**. Body armour **100** may provide protection for other parts of the body, such as the shins, knees, thighs, forearms, elbows, upper arms, shoulders, chest and back. Individual items of body armour may be provided to protect individual body parts (as shown in FIG. **9**), or alternatively may be combined in apparel comprising multiple armoured regions **101** to protect more than one body part (as shown in FIG. **10**). Such body armour **100** may be worn for the same activities as helmets, discussed above, including for combat, sports, and motorcycling.

[0129] The body armour **100** may have the same layered structure as the example helmets described above. For

example, the body armour **100** may comprise an outer shell **2** as described above, one or more energy absorbing layers **3** as described above, and/or an interface layer as described above, and/or layers that move relative to each other as described above, and/or layers may be connected by connectors **5** as described above.

[0130] Connectors that may be used within a helmet are described below. It should be appreciated that these connectors may be used in a variety of contexts and are not to be limited to use within helmets. For example, they may be used in other apparatuses that provide impact protection, such as body armour or padding for sports equipment.

[0131] It should be appreciated that the connectors may be used for connecting any two parts of an apparatus together. In the context of helmets, it should be appreciated in particular that the connectors may be used for connecting any two parts of helmets such as those discussed above that are configured to move relative to each other.

[0132] Where a connector is described as having a first part connected to a first part of an apparatus and a second part connected to a second part of an apparatus, it should be appreciated that, with suitable modifications, this may be reversed. It should also be appreciated that where an apparatus has first and second parts connected by plural connectors, the plural connectors need not have the same configuration as each other.

[0133] FIG. **11** shows a first side of an example connector **20**. As shown, the connector **20** comprises a first attachment part **21**, a first low friction layer **22** and a deformable layer **23**.

[0134] As shown in FIG. **11**, the first attachment part **21** is arranged on the first side of the connector **20**. The first attachment part **21** is configured to be attached to a first part of an apparatus, such that the first side of the connector **20** faces the first part of the apparatus. The direction in which the first side of the connector **20** faces, is referred to as the first direction. In certain embodiment the first direction may be a direction substantial perpendicular to the connector **20**, and/or the first part of the apparatus, and/or the second part of the apparatus. In an example in which the apparatus is a helmet, this may be a radial direction of the helmet, when the connector **20** is in use.

[0135] As shown in FIG. **11**, the first low friction layer **22** is arranged on the first side of the connector **20**. The first low friction layer **22** is configured to reduce friction between the connector **20** and the first part of the apparatus.

[0136] As shown in FIG. **11**, the deformable layer **23** is provided, at least, between the first attachment part **21** and the first low friction layer **22**. As shown, the deformable layer **23** extends in a direction substantially perpendicular to the first direction defined above. The deformable layer **23** is formed from a deformable material and is configured to allow the first attachment part **21** and the first low friction layer **22** to move relative to each other at least in a direction substantially perpendicular to the first direction. This movement is illustrated by the arrows in FIG. **11**. In certain embodiment, the direction of extension and deformation of the deformable layer may be substantially parallel to the connector **20**, and/or the first part of the apparatus, and/or the second part of the apparatus. In an example in which the apparatus is a helmet, this may be a circumferential direction of the helmet, when the connector **20** is in use.

[0137] FIG. **12** shows a second side of the example connector **20** shown in FIG. **11**. As shown, the connector **20**

may further comprise a second attachment part **24** and a second low friction layer **25**. As shown, the deformable layer **23** is also provided, at least, between the second attachment part **24** and the second low friction layer **25**. As illustrated by the arrows in FIG. 12, the deformable layer is also configured to allow the second attachment part **24** and second low friction layer **25** to move relative to each other.

[0138] As shown in FIG. 12, the second attachment part **24** may be arranged on the second side of the connector **20**. The second attachment part **24** is configured to be attached to a second part of an apparatus, such that the second side of the connector **20** faces the second part of the apparatus.

[0139] As shown in FIG. 12, the second low friction layer **25** may be arranged on the second side of the connector **20**. The second low friction layer **25** is configured to reduce friction between the connector **20** and the second part of the apparatus.

[0140] FIG. 13 shows a cross-section through the example connector **20** shown in FIGS. 11 and 12. As shown, the first attachment part **21** may be provided on a first side of the deformable layer **23**. As shown, the first low friction layer **22** may be provided on the first side of the deformable layer **23**. As shown, the second attachment part **24** may be provided on a second side of the deformable layer **23**, opposite the first side. As shown, the second low friction layer **25** may be provided on the second side of the deformable layer **23**.

[0141] As shown in FIG. 11, the first low friction layer **22** may surround the first attachment part **21**. For example, as shown, the first low friction layer **22** may be annular in shape. For example, as shown, the first attachment part **21** may be circular in shape. However, in other examples, alternative shapes may be used. In other examples, the first low friction layer **22** may partially surround the first attachment part **21**. As shown in FIG. 11, the first low friction layer **22** may be provided at an edge of the connector **20** and the first attachment part **21** may be provided at a centre of the connector **20**.

[0142] As shown in FIG. 13, the second attachment part **24** may be arranged to substantially cover a portion of the second side of the connector opposite a portion of the first side of the connector **20** at which the first low friction layer **22** is arranged. As shown in FIGS. 11 and 12, the second attachment part **24** may be substantially the same shape as the first low friction layer **22**.

[0143] As shown, the second low friction layer **25** may be arranged to substantially cover a portion of the second side of the connector opposite a portion of the first side of the connector **20** at which the first attachment part **21** is arranged. As shown in FIGS. 11 and 12, the second low friction layer **25** may be substantially the same shape as the first attachment part **21**.

[0144] As shown in FIG. 12, the second attachment part **24** may surround the second low friction layer **25**. For example, as shown, the second low friction layer **25** may be circular in shape. For example, as shown, the second attachment part **21** may be annular in shape. However, in other examples, alternative shapes may be used. In other examples, the second attachment part **24** may partially surround the second low friction layer **25**. As shown in FIG. 12, the second low friction layer **25** may be provided the centre of the connector **20** and the second attachment part **24** may be provided at the edge of the connector **20**.

[0145] As shown in FIGS. 11 to 13, the first attachment part **21** and the second attachment part **24** may be displaced

from each other in a direction along the deformable material **23**. This may allow movement of the first part of the apparatus connected to the first attachment part **21** and the second part of the apparatus connected to the second attachment part **24**. The displacement between the first attachment part **21** and the second attachment part **24** may be at least 1 centimetre to allow at least 1 centimetre of relative movement. The deformable material **23** may have a diameter (or other dimension, if not circularly shaped) between around 15 millimetres and 50 millimetres, for example.

[0146] As shown in FIG. 13, the deformable layer **23** may be formed from a sheet of the deformable material. This may be a sheet of stretchable fabric, for example. One or more of the first attachment part **21**, the first low friction layer **22**, the second attachment part **24** and the second low friction layer **25** may be attached to the deformable layer **23** by means of high frequency welding, adhesive, or otherwise.

[0147] The deformable material **23** may be configured to stretch at least 1 cm, in normal use, so as to provide at least 1 cm relative movement between the first attachment part **21** and the second attachment part **24**. The deformable material **23** may stretch in to two orthogonal directions.

[0148] As shown in FIG. 13, the first low friction layer **22** may be formed from a sheet of low friction material. For example, the first low friction layer may be formed from a thin layer of plastic, as in the present example, such as polycarbonate (PC) or thermoplastic polyurethane (TPU). The first low friction layer **22** may be no more than 0.5 millimetre's thick, or preferably no more than 0.1 mm thick, for example. The low friction layer **22** may be flexible, e.g. such that it is able to bend. Alternatively, the first low friction layer **22** may be formed from a layer of brushed nylon. Alternatively, the first low friction layer **22** may be formed from any of the low friction materials described above in relation to providing low friction between layers of an apparatus.

[0149] As shown in FIG. 13, the second low friction layer **25** may be formed from a sheet of low friction material. For example, the second low friction layer **25** may be formed from a layer of brushed nylon, as in the present example. Alternatively, the second low friction layer **25** may be formed from a thin layer of plastic as described above. Alternatively, the first low friction layer **22** may be formed from any of the low friction materials described above in relation to providing low friction between layers of an apparatus.

[0150] As shown in FIG. 13, the first attachment part **21** may be formed from a sheet of material. For example, the first attachment part **21** may be formed from a hook and loop material, such as Velcro™, as in this example. The first attachment part **21** may be formed from the hook part of a hook and loop material, the loop part, or a combination of both. As shown in FIG. 11, the first attachment part **21** may be disk shaped, e.g. formed from a disk of a hook and loop material. Alternatively, the first attachment part **21** may be formed from a layer of adhesive material, such as double-sided adhesive tape.

[0151] As shown in FIG. 13, the second attachment part **24** may be formed from a sheet of material. For example, the second attachment part **24** may be formed from a layer of adhesive material, such as double-sided adhesive tape, as in this example. Alternatively, the first attachment part **21** may be formed from a layer of a hook and loop material as described above.

[0152] FIG. 14 shows the example connector in use in an apparatus—in this example, a helmet. As shown, the helmet comprises an energy absorbing layer 3, a low-friction layer 4 and a liner 15 that is connected to the rest of the helmet by the connector 20. The low friction layer 4 of the helmet, in this example, is a strip, shown in cross-section in FIG. 14. As shown, the connector 20 may be provided over the low friction layer 4. As shown, the first attachment part 21, in this example a hook and loop attachment part, attaches the connector 20 to the liner 15. As shown, the second attachment part 24, in this example an adhesive layer, attaches the connector 20 to the energy absorbing layer 3 and the low friction layer 4 of the helmet. As shown, the connector 20 bends over the low friction layer 4 of the helmet. As shown, the connector 20 may attach to the low friction layer 4 (on the energy absorbing layer 3) and also a portion of the energy absorbing layer 3 adjacent the low friction layer 4, e.g. on two opposite sides of the low friction layer 4. As shown the first and second low friction layers 22, 25 respectively interface with opposing surfaces of the liner 15 and the energy absorbing layer.

[0153] In an alternative example, the connector 2 may bend over a portion of the energy absorbing layer 3 that protrudes from the rest of the energy absorbing layer. In such an example, the low friction layer 4 shown in FIG. 14 may or may not be provided on the protruding portion of the energy absorbing layer 3.

[0154] FIG. 15 shows a second example connector 20 in use in a second example apparatus—in this example, a helmet. As shown, the helmet comprises an energy absorbing layer 3 and a liner 15. As shown, the first attachment part 21, in this example a hook and loop attachment part, attaches the connector 20 to the liner 15. As shown, the second attachment part 24, in this example also a hook and loop attachment part, attaches the connector 20 to the energy absorbing layer 3. As shown, the energy absorbing layer 3 may comprise a recess at which the second attachment part 24 is attached. In this example the recess is circular. As shown, a corresponding hook and loop attachment part 24A is provided in the recess and is fixed to the energy absorbing layer 3, e.g. by an adhesive. As shown the first and second low friction layers 22, 25 respectively interface with opposing surfaces of the liner 15 and the energy absorbing layer.

[0155] In an alternative example, the first attachment part and/or the second attachment part may be double-sided adhesive tape instead of hook and loop attachment parts.

[0156] Helmets as described above may be used in various activities. These activities include combat and industrial purposes, such as protective helmets for soldiers and hard-hats or helmets used by builders, mine-workers, or operators of industrial machinery for example.

[0157] Helmets, are also common in sporting activities. For example, protective helmets may be used in ice hockey, cycling, motorcycling, motor-car racing, skiing, snowboarding, skating, skateboarding, equestrian activities, American football, baseball, rugby, soccer, cricket, lacrosse, climbing, golf, airsoft, roller derby and paintballing.

[0158] Examples of injuries that may be prevented or mitigated by the helmets described above include Mild Traumatic Brain Injuries (MTBI) such as concussion, and Severe Traumatic Brain Injuries (STBI) such as subdural haematomas (SDH), bleeding as a consequence of blood vessels rupturing, and diffuse axonal injuries (DAI), which

can be summarized as nerve fibres being over stretched as a consequence of high shear deformations in the brain tissue.

[0159] Depending on the characteristics of the rotational component of an impact, such as the duration, amplitude and rate of increase, either concussion, SDH, DAI or a combination of these injuries can be suffered. Generally speaking, SDH occur in the case of accelerations of short duration and great amplitude, while DAI occur in the case of longer and more widespread acceleration loads.

[0160] Variations of the above described examples are possible in light of the above teachings. It is to be understood that the invention may be practiced otherwise and specifically described herein without departing from the spirit and scope of the invention.

1. A connector for connecting first and second parts of an apparatus, the connector comprising:

- a first attachment part arranged on a first side of the connector, configured to be attached to the first part of the apparatus such that the first side of the connector faces in a first direction towards the first part of the apparatus;
- a first low friction layer arranged on the first side of the connector and configured to reduce friction between the connector and the first part of the apparatus;
- a deformable layer formed from a deformable material, provided at least between the first attachment part and the first low friction layer extending in a direction substantially perpendicularly to the first direction and configured to allow the first attachment part and first low friction layer to move relative to each other in a direction substantially perpendicularly to the first direction.

2. The connector of claim 1, wherein the deformable layer is formed from a sheet of the deformable material.

3. The connector of claim 2, wherein the deformable material is a sheet of stretchable fabric.

4. The connector of claim 2, wherein the first attachment part is arranged on a first side of the sheet of deformable material.

5. The connector of claim 2, wherein the first low friction layer is arranged on the first side of the sheet of deformable material.

6. The connector of claim 1, wherein the low friction layer is formed from a sheet of low friction material.

7. The connector of claim 1, wherein the first low friction layer is flexible.

8. The connector of claim 1, wherein the first low friction layer is formed from a thin layer of plastic, optionally wherein the plastic is polycarbonate.

9. (canceled)

10. The connector of claim 1, wherein the first low friction layer is no more than 0.5 mm thick.

11. The connector of claim 1, wherein the low friction layer surrounds the first attachment part, optionally wherein the low friction layer is annular.

12. (canceled)

13. The connector of claim 1, wherein the first attachment part comprises a hook and loop material, optionally wherein the first attachment part comprises a disk of hook and loop material.

14. (canceled)

15. The connector of claim **1**, further comprising a second attachment part on a second side of the connector opposite the first side, configured to be attached to the second part of the apparatus.

16. The connector of claim **15**, wherein the first and second attachment parts are displaced from each other in a direction along the deformable material, optionally wherein the deformable material is configured to stretch by at least 1 cm, so as to provide at least 1 cm relative movement between the first attachment part and the second attachment part.

17. (canceled)

18. The connector of claim **15**, wherein the second attachment part is arranged to substantially cover a portion of the second side of the connector opposite a portion of the first side of the connector at which the first low friction layer is arranged.

19. The connector of claim **1**, further comprising a second low friction layer on a second side of the connector opposite the first side, configured to reduce friction between the connector and the second part of the apparatus, optionally wherein the second low friction layer is arranged to sub-

stantially cover a portion of the second side of the connector opposite a portion of the first side of the connector at which the first attachment part is arranged.

20. (canceled)

21. An apparatus comprising:

a first part;

a second part; and

at least one connector according to claim **1** connecting the first and second parts.

22. The apparatus of claim **22**, wherein the second part of the apparatus comprises a low friction layer arranged to face the one or more connectors.

23. The apparatus of claim **22**, wherein the apparatus is protective apparel and the first part of the apparatus comprises an interface layer configured to interface with a wearer.

24. The apparatus of claim **22**, wherein the second part comprises at least one of a hard shell and an energy absorbing layer.

25. The apparatus of claim **22**, wherein the apparatus is a helmet.

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